CENG381 Stochastic Processes

Wald's Equality and De Moivre's Martingale - Answer Key 1

Q1 Solution

a) E[X_i] and Var[X_i] for a fair die:

$E[X] = (1+2+3+4+5+6)/6 = 21/6 = 7/2 = 3.5$

$E[X^2] = (1+4+9+16+25+36)/6 = 91/6 \sim 15.167$

$Var[X] = E[X^2] - (E[X])^2 = 91/6 - (7/2)^2$

$= 91/6 - 49/4$

$= 182/12 - 147/12$

$= 35/12 \sim 2.917$

b) Approximate E[T] via Wald's Equality:

Wald's Equality: $E[S(T)] = E[X] * E[T]$

Since $S(T) >= 100$ and the overshoot is small relative to 100,
we approximate $E[S(T)] \sim 100$. Therefore:

$100 \sim E[X] * E[T] = (7/2) * E[T]$

$E[T] \sim 100 / (7/2) = 200/7 \sim 28.57$ rolls

You need approximately 28-29 die rolls to reach a sum of 100.

c) Probability of hitting exactly 100:

By renewal theory (the renewal process where $S(n)$ is the position and 100 is a renewal target), the long-run probability of the sum passing through any specific integer m is approximately $1/E[X] = 2/7$.

$P(S(n) = 100 \text{ for some } n) \sim 1/E[X] = 2/7 \sim 0.286$

About a 28.6% chance of landing exactly on 100 at some step.
The remaining ~71.4% of the time, the sum jumps over 100.
(For example, at sum 95 you roll a 6 and land on 101.)

Q2 Solution

a) Verify r = q/p and E[r^{X_{n+1}}] = 1:

$p = 3/5$, $q = 2/5$, $r = q/p = (2/5)/(3/5) = 2/3$

Verify $E[r^{X_{n+1}}] = 1$:

$$E[r^{X_{n+1}}] = p * r^{+1} + q * r^{-1}$$
$$= (3/5) * (2/3) + (2/5) * (3/2)$$
$$= 2/5 + 3/5$$
$$= 1 \quad \text{(OK)}$$

Therefore:

$$E[M(n+1) \mid F_n] = M(n) * E[r^{X_{n+1}}] = M(n) * 1 = M(n)$$

$M(n) = r^{S(n)} = (2/3)^{S(n)}$ is a martingale.

b) Gambler's Ruin probability h via OST on De Moivre's martingale:

Apply OST to $M(n) = (2/3)^{S(n)}$ with T = first hit of $\{0, 6\}$, starting at $S(0) = k = 2$:

$$E[M(T)] = E[M(0)] = r^k = (2/3)^2 = 4/9$$

At stopping time T:

$M(T) = r^6 = (2/3)^6 = 64/729$ with probability h (reach N=6)

$M(T) = r^0 = 1$ with probability 1-h (reach 0)

Setting up the equation:

$$(64/729) * h + 1 * (1 - h) = 4/9$$
$$(64/729) * h + 1 - h = 4/9$$
$$h * (64/729 - 1) = 4/9 - 1$$
$$h * (64/729 - 729/729) = 4/9 - 9/9$$
$$h * (-665/729) = -5/9$$
$$h = (5/9) / (665/729)$$
$$= (5/9) * (729/665)$$
$$= 5 * 81 / 665$$
$$= 405 / 665$$
$$= 81/133 \sim 0.609$$

c) Comparison to symmetric walk and P(ruin):

Symmetric walk ($p = 1/2$): $h_{sym} = k/N = 2/6 = 1/3 \sim 0.333$

Biased walk (p = 3/5): h = 81/133 ~ 0.609

With upward bias (p = 3/5 > 1/2), the probability of reaching the goal INCREASES significantly: from 33.3% to 60.9%.

Intuition: with drift +1 on average per step (E[X] = 2p-1 = 0.2 > 0), the walk tends to move toward higher states, making N=6 more reachable than ruin at 0 from position k=2.

P(ruin before goal) = 1 - h = 1 - 81/133 = 52/133 ~ 0.391

Q3 Solution

a) Risk-neutral probability p*:

Write S(1) = S(0) * Y_1 where Y_1 = u=1.1 with prob p or d=0.9 with prob 1-p.

For S(n) to be a martingale (E[S(1)] = S(0)):

E[S(1)] = S(0) * E[Y_1] = S(0) * (p*u + (1-p)*d) = S(0)
=> p*u + (1-p)*d = 1
=> p*1.1 + (1-p)*0.9 = 1
=> 0.9 + p*(1.1 - 0.9) = 1
=> 0.9 + 0.2*p = 1
=> p* = 0.1/0.2 = 0.5

The risk-neutral probability is p* = 1/2 (equal weight to up and down).
This makes sense: when p*u + (1-p*)d = 1, the expected return is zero,
meaning the asset has no drift under this measure.

b) S(n) is a martingale under p* = 1/2:

Write S(n+1) = S(n) * Y_{n+1} where Y_{n+1} = 1.1 (prob 1/2) or 0.9 (prob 1/2),
and Y_{n+1} is independent of F_n.

E[S(n+1) | F_n] = S(n) * E[Y_{n+1} | F_n]
= S(n) * E[Y_{n+1}] (by independence)
= S(n) * (0.5*1.1 + 0.5*0.9)
= S(n) * (0.55 + 0.45)

$= S(n) * 1.0$

$= S(n)$ (OK)

So $S(n)$ is indeed a martingale under $p^* = 1/2$.

c) Jensen's Inequality and real-world measure:

Under the real-world measure with $p > p^* = 1/2$ (upward drift):

$E[Y] = p*1.1 + (1-p)*0.9 = 0.9 + 0.2*p > 0.9 + 0.2*0.5 = 1$

(since $p > 0.5$ implies $0.2*p > 0.1$)

So $E[Y] > 1$ under the real-world measure with $p > 1/2$.

$E[S(n+1) \mid F_n] = S(n) * E[Y] > S(n)$

=> $S(n)$ is a SUBMARTINGALE under the real-world measure with upward drift.

Jensen's connection:

The multiplicative model $S(n) = S(0) * Y_1 * \ldots * Y_n$ can be written as $S(n) = S(0) * e^{Z_1 + \ldots + Z_n}$ where $Z_i = \log(Y_i)$. By Jensen's inequality with the convex function $f(x) = e^x$:

$E[S(n)] = S(0)*E[e^{Z_1+\ldots+Z_n}] >= S(0)*e^{E[Z_1+\ldots+Z_n]}$

Since $E[Z_i] = p*\log(1.1) + (1-p)*\log(0.9) > 0$ when $p > 1/2$ (more up moves on average, and $\log(1.1) > |\log(0.9)|$ is not exactly true but the principle holds for $E[Y]>1$), the asset price grows on average.

This confirms $E[S(n)] > S(0)$ under upward drift, i.e. $S(n)$ is a submartingale.